```
def forward(self,x):
    x = F.normalize(x,dim=0)
    y = F.relu(self.l1(x))
    y = F.relu(self.l2(y))
    actor = F.log_softmax(self.actor_lin1(y),dim=0)   3
    c = F.relu(self.l3(y.detach()))
    critic = torch.tanh(self.critic_lin1(c))          4
    return actor, critic                              5
```

- ***1*** PyTorch wraps Python's built-in multiprocessing library, and the API is the same.
- ***2*** Defines a single combined model for the actor and critic
- ***3*** The actor head returns the log probabilities over the 2 actions.
- ***4*** The critic returns a single number bounded by –1 and +1.
- ***5*** Returns the actor and critic results as a tuple

For CartPole, we have a fairly simple neural network, apart from having two output heads. In listing 5.4 we first normalize the input so that the state values are all within the same range; then the normalized input is fed through the first two layers, which are ordinary linear layers with the ReLU activation functions. Then we fork the model into two paths.

The first path is the actor head that takes the output of layer 2 and applies another linear layer and then the `log_softmax` function. The `log_softmax` is logically equivalent to doing `log(softmax(...))`, but the combined function is more numerically stable because if you compute the functions separately you might end up with overflowed or underflowed probabilities after the `softmax`.

The second path is the critic head, which applies a linear layer and ReLU to the output of layer 2, but notice that we call `y.detach()`, which detaches the `y` node from the graph so the critic's loss won't backpropagate and modify the weights in layers 1 and 2 (figure 5.11). Only the actor will cause these weights to be modified. This prevents conflict between what the actor and critic want when the actor and critic are trying to make opposing updates to the earlier layers. With two-headed models, it often makes sense to make one head dominant and allow it to control most of the parameters by detaching the other head from the first several layers. Lastly, the critic applies another linear layer with the tanh activation function that bounds the